4.1.2 Alkanes

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Properties of alkanes		
(a)	alkanes as saturated hydrocarbons containing single C–C and C–H bonds as σ-bonds (overlap of orbitals directly between the bonding atoms); free rotation of the σ-bond	Hybridisation **not** required. HSW1 Use of model of orbital overlap to explain covalent bonding in organic compounds.
(b)	explanation of the tetrahedral shape and bond angle around each carbon atom in alkanes in terms of electron pair repulsion **(see also 2.2.2 g–h)**	*M4.1, M4.2* Learners should be able to draw 3-D diagrams.
(c)	explanation of the variations in boiling points of alkanes with different carbon-chain length and branching, in terms of induced dipole–dipole interactions (London forces) **(see also 2.2.2 k)**	*M3.1*
Reactions of alkanes		
(d)	the low reactivity of alkanes with many reagents in terms of the high bond enthalpy and very low polarity of the σ-bonds present **(see also 2.2.2 j)**	HSW1 Use of ideas about enthalpy and polarity to explain macroscopic properties of alkanes.
(e)	complete combustion of alkanes, as used in fuels, and the incomplete combustion of alkane fuels in a limited supply of oxygen with the resulting potential dangers from CO	
(f)	the reaction of alkanes with chlorine and bromine by radical substitution using ultraviolet radiation, including a mechanism involving homolytic fission and radical reactions in terms of initiation, propagation and termination **(see also 4.1.1 f–g)**	Learners are **not** required to use 'half curly arrows' in this mechanism. Equations should show which species are radicals using a single 'dot', •, to represent the unpaired electron.
(g)	the limitations of radical substitution in synthesis by the formation of a mixture of organic products, in terms of further substitution and reactions at different positions in a carbon chain.	